

Nicklas Hansen

hello@nicklashansen.com | nicklashansen.com | [nicklashansen](https://github.com/nicklashansen) | [ncklas](https://www.linkedin.com/in/ncklas) | [New York City, NY](#)

Experience

Courant Institute, New York University

Postdoctoral Associate

· Advised by Lerrel Pinto and Yann LeCun.

New York City, NY

Aug 2026 - now

NVIDIA Research

Research Intern

· Mentored by Yashraj Narang and Dieter Fox.

Seattle, WA

June 2024 - Feb 2025

Meta AI (FAIR)

Research Scientist Intern

· Mentored by Aravind Rajeswaran.

Menlo Park, CA

June 2022 - Dec 2022

raffle.ai

Machine Learning Intern

· I built and open-sourced a cross-domain text-to-SQL model in PyTorch.

Copenhagen, Denmark

June 2019 - Aug 2019

Education

University of California, San Diego

PhD student, Computer Science and Engineering, GPA: 3.85/4.0

· Advised by Xiaolong Wang and Hao Su. NVIDIA Graduate Fellow.

San Diego, CA

Sep 2021 - June 2026

University of California, Berkeley

Visiting Student, GPA: 4.0/4.0

· Spar Nord Fonden's FinTech scholarship recipient. SCET's Collider Cup finalist.

Berkeley, CA

Spring 2020

Technical University of Denmark

MS Mathematical Modeling & Computation, GPA: 11.2/12.0

· Special topics in machine learning. Advised by Ole Winther.

Kongens Lyngby, Denmark

Feb 2019 - Feb 2021

Technical University of Denmark

BS Software Technology, GPA: 8.2/12.0

· **Nanyang Technological University, Singapore** - semester abroad, Fall 2017.

Kongens Lyngby, Denmark

Sep 2015 - Dec 2018

Publications & Preprints

Catch It All!: Generalizable Dynamic Catching with a Dexterous Hand

Paper under review

Arth Shukla*, Kehlani Fay*, Huaze Liu[^], Darin Djabri[^], Alexiy Buynitsky, **Nicklas Hansen**

preprint available soon

Under review

2026

SNAP: One-Shot Dexterous Manipulation via Functional Grasping Policies

Paper under review

Alexiy Buynitsky, **Nicklas Hansen**, Carmelo Sferrazza, Jianglong Ye, Xiaolong Wang

preprint available soon

Under review

2026

Causal World-Action Models for Failure-Aware Robot Policies

Paper under review

Quanquan Peng, Yutong Liang, Rui Yan, **Nicklas Hansen**, Xiaolong Wang

preprint available soon

Under review

2026

Simulation Priors for Rapid Adaptation in Model-Based RL for Autonomous Racing

Paper under review

Adrian Remonda, Jiajun Xi, **Nicklas Hansen**, Marcus Greiff, Thomas Lew, John Talbot, John Subosits, Xiaolong Wang

preprint available soon

Under review

2026

HiGym: Learning to Act in High-Dimensional Spaces

Paper under review

Harshit Dhankhar, Xiaolong Wang, **Nicklas Hansen**

preprint available soon

Under review

2026

Generating Robot Hands from Human Demonstrations

arXiv preprint

Yi Sha, **Nicklas Hansen**, Xueqian Bai, Carmelo Sferrazza, Michael T. Tolley, Xiaolong Wang

<https://arxiv.org/abs/2606.20549>

Preprint

2026

Hallucination in World Models is Predictable and Preventable

arXiv preprint

Nicklas Hansen, Xiaolong Wang

<https://arxiv.org/abs/2606.27326>

Preprint

2026

Bayesian Curriculum Learning over Latent Geometry in Large Language Models

arXiv preprint

Darrien McKenzie, **Nicklas Hansen**, Xiaolong Wang

<https://arxiv.org/abs/2606.19750>

Preprint

2026

Abstracting Task-Centric World Models from Visual Foundations

arXiv preprint

Minghao Fu, Fan Feng, **Nicklas Hansen**, Biwei Huang

<https://arxiv.org/abs/2605.25620>

Preprint

2026

Learning Reasoning-Guided Robot Policies via Task-Aligned Simulated Futures

European Conference on Computer Vision (**ECCV**)

Hanan Gani, Tejal Kulkarni, Madhoolika Venkata Chodavarapu, **Nicklas Hansen**, Manmohan Chandraker

<https://arxiv.org/abs/2607.06018>

2026

Generalizable Robotic Insertion with World Models

International Conference on Intelligent Robots and Systems (**IROS**)

Nicklas Hansen, Iretiayo Akinola, Yijie Guo, Jie Xu, Bingjie Tang, Hao Su, Xiaolong Wang, Abhishek Gupta, Dieter Fox, Yashraj Narang

<https://openreview.net/pdf?id=DR3n6lqGKI>

2026

TIPS: Turn-Level Information-Potential Reward Shaping for Search-Augmented LLMs

International Conference on Learning Representations (**ICLR**)

Yutao Xie, Nathaniel Thomas, **Nicklas Hansen**, Yang Fu, Li Erran Li, Xiaolong Wang

<https://arxiv.org/abs/2603.22293>

2026

Learning Massively Multitask World Models for Continuous Control

International Conference on Learning Representations (**ICLR**)

Nicklas Hansen, Hao Su*, Xiaolong Wang*

<https://arxiv.org/abs/2511.19584>

2026

Learning to Design Soft Hands using Reward Models

International Conference on Robotics and Automation (ICRA)

2025

Xueqian Bai, **Nicklas Hansen**, Adabhav Singh, Michael T. Tolley, Yan Duan, Pieter Abbeel, Xiaolong Wang, Sha Yi

<https://arxiv.org/abs/2510.17086>

MyoChallenge 2024: A New Benchmark for Physiological Dexterity and Agility in Bionic Humans

2025

Conference on Neural Information Processing Systems (NeurIPS)

Cheryl Wang, Chun Kwang Tan, [...], **Nicklas Hansen**, [...], Vikash Kumar, Vittorio Caggiano (26 authors)

<https://openreview.net/forum?id=1dSLbhErNv>

Multi-Stage Manipulation with Demonstration-Augmented Reward, Policy, and World Model Learning

2025

International Conference on Machine Learning (ICML)

Adrià López Escoriza, **Nicklas Hansen**, Stone Tao, Tongzhou Mu, Hao Su

<https://arxiv.org/abs/2503.01837>

Policy Learning with Multi-Task World Models

International Conference on Learning Representations (ICLR)

2025

Ignat Georgiev, Varun Giridhar, **Nicklas Hansen**, Animesh Garg

<https://arxiv.org/abs/2407.02466>

Hierarchical World Models as Visual Whole-Body Humanoid Controllers

International Conference on Learning Representations (ICLR)

2025

Nicklas Hansen, Jyothir S V, Vlad Sobal, Yann LeCun, Xiaolong Wang*, Hao Su*

<https://arxiv.org/abs/2405.18418>

A Simulation Benchmark for Autonomous Racing with Large-Scale Human Data

Conference on Neural Information Processing Systems (NeurIPS)

2024

Adrian Remonda, **Nicklas Hansen**, Ayoub Raji, Nicola Musiu, Marko Bertogna, Eduardo E. Veas, Xiaolong Wang

<https://arxiv.org/abs/2407.16680>

A Recipe for Unbounded Data Augmentation in Visual Reinforcement Learning

Reinforcement Learning Conference (RLC)

2024

Abdulaziz Almuzairee, **Nicklas Hansen**, Henrik I. Christensen

<https://arxiv.org/abs/2405.17416>

TD-MPC2: Scalable, Robust World Models for Continuous Control

Spotlight

International Conference on Learning Representations (ICLR)

2024

Nicklas Hansen, Hao Su*, Xiaolong Wang*

<https://arxiv.org/abs/2310.16828>

Open X-Embodiment: Robotic Learning Datasets and RT-X Models

Best Conference Paper

International Conference on Robotics and Automation (ICRA)

2024

Open X-Embodiment Collaboration, [...], **Nicklas Hansen**, [...] (173 authors)

<https://arxiv.org/abs/2310.08864>

MoDem-V2: Visuo-Motor World Models for Real-World Robot Learning

International Conference on Robotics and Automation (ICRA)

2024

Patrick Lancaster, **Nicklas Hansen**, Aravind Rajeswaran, Vikash Kumar

<https://arxiv.org/abs/2309.14236>

Finetuning Offline World Models in the Real World

Oral

Conference on Robot Learning (CoRL)

2023

Yunhai Feng*, **Nicklas Hansen***, Ziyang Xiong*, Chandramouli Rajagopalan, Xiaolong Wang

<https://arxiv.org/abs/2310.16029>

Multi-Task Real Robot Learning with Generalizable Neural Feature Fields Conference on Robot Learning (CoRL) Yanjie Ze, Ge Yan, Yueh-Hua Wu, Annabella Macaluso, Yuying Ge, Jianglong Ye, Nicklas Hansen , Li Erran Li, Xiaolong Wang https://arxiv.org/abs/2308.16891	Oral 2023
On Pre-Training for Visuo-Motor Control: Revisiting a Learning-from-Scratch Baseline International Conference on Machine Learning (ICML) Nicklas Hansen* , Zhechen Yuan*, Yanjie Ze*, Tongzhou Mu*, Aravind Rajeswaran^, Hao Su^, Huazhe Xu^, Xiaolong Wang^ https://arxiv.org/abs/2212.05749	2023
MoDem: Accelerating Visual Model-Based Reinforcement Learning with Demonstrations International Conference on Learning Representations (ICLR) Nicklas Hansen , Yixin Lin, Hao Su, Xiaolong Wang, Vikash Kumar, Aravind Rajeswaran https://arxiv.org/abs/2212.05698	2023
On the Feasibility of Cross-Task Transfer with Model-Based Reinforcement Learning International Conference on Learning Representations (ICLR) Yifan Xu*, Nicklas Hansen* , Zirui Wang, Yung-Chieh Chan, Hao Su, Zhouwen Tu https://arxiv.org/abs/2210.10763	2023
Visual Reinforcement Learning with Self-Supervised 3D Representations IEEE Robotics and Automation Letters (RA-L) Yanjie Ze*, Nicklas Hansen* , Yinbo Chen, Mohit Jain, Xiaolong Wang https://arxiv.org/abs/2210.07241	2023
Graph Inverse Reinforcement Learning from Diverse Videos Conference on Robot Learning (CoRL) Sateesh Kumar, Jonathan Zamora*, Nicklas Hansen* , Rishabh Jangir, Xiaolong Wang https://arxiv.org/abs/2207.14299	Oral 2022
Temporal Difference Learning for Model Predictive Control International Conference on Machine Learning (ICML) Nicklas Hansen , Xiaolong Wang*, Hao Su* https://arxiv.org/abs/2203.04955	Spotlight 2022
Look Closer: Bridging Egocentric and Third-Person Views with Transformers for Robotic Manipulation IEEE Robotics and Automation Letters (RA-L) Rishabh Jangir*, Nicklas Hansen* , Sambaran Ghosal, Mohit Jain, Xiaolong Wang https://arxiv.org/abs/2201.07779	2022
Learning Vision-Guided Quadrupedal Locomotion End-to-End with Cross-Modal Transformers International Conference on Learning Representations (ICLR) Ruihan Yang*, Minghao Zhang*, Nicklas Hansen , Huazhe Xu, Xiaolong Wang https://arxiv.org/abs/2107.03996	Spotlight 2022
Stabilizing Deep Q-Learning with ConvNets and Vision Transformers under Data Augmentation Conference on Neural Information Processing Systems (NeurIPS) Nicklas Hansen , Hao Su, Xiaolong Wang https://arxiv.org/abs/2107.00644	2021

Generalization in Reinforcement Learning by Soft Data Augmentation

International Conference on Robotics and Automation (ICRA)

2021

Nicklas Hansen, Xiaolong Wang

<https://arxiv.org/abs/2011.13389>

Self-Supervised Policy Adaptation during Deployment

International Conference on Learning Representations (ICLR)

Spotlight

2021

Nicklas Hansen, Rishabh Jangir, Yu Sun, Guillem Alenyà, Pieter Abbeel, Alexei A. Efros,

Lerrel Pinto, Xiaolong Wang

<https://arxiv.org/abs/2007.04309>

Short Term Blood Glucose Prediction Based on Continuous Glucose Monitoring Data

IEEE Engineering in Medicine and Biology Conference (EMBC)

2020

Ali Mohebbi, Alexander R. Johansen, Nicklas Hansen, Peter E. Christensen, Jens M. Tarp,

Morten L. Jensen, Henrik Bengtsson, Morten Mørup

<https://arxiv.org/abs/2002.02805>

Teaching

University of California, San Diego

ECE285 Introduction to Visual Learning

· Held office hours, graded assignments.

Teaching Assistant

Spring 2024

Technical University of Denmark

Reinforcement Learning

· Special course that I co-organized w/ Prof. Ole Winther for a group of students.

Three weeks of full-time study.

Co-organizer

Jan 2021

Technical University of Denmark

02456 Deep Learning

· Significant course material contributions, supervised 100+ students' projects on reinforcement learning.

02454 Introduction to Cognitive Science

· Assisted tutorial sessions, graded assignments.

Teaching Assistant

Fall 2019, Fall 2020

Fall 2019

Current and Former Mentees

Rishabh Jangir (MS, UCSD → ML Engineer, Skild AI)

2020 - 2022

Mohit Jain (MS, UCSD → ML Engineer, Pinterest)

2020 - 2022

Xinyue Chen (BS, NYU Shanghai → PhD, UC Berkeley)

2021 - 2022

Sateesh Kumar (MS, UCSD → PhD, UT Austin)

2021 - 2023

Jonathan Zamora-Anaya (BS, UCSD → MS, USC)

2021 - 2023

Sambaran Ghosal (MS, UCSD → ML Engineer, BrainCorp)

2021 - 2023

Zirui "Colin" Wang (BS, UCSD → PhD, UC Berkeley)

2022 - 2023

Ziyan Xiong (BS, Tsinghua University → PhD, UMass)

2022 - 2023

Yanjie Ze (BS, SJTU → PhD, Stanford)

2021 - 2023

Yunhai Feng (MS, UCSD → PhD, Cornell)

2022 - 2024

Chandramouli Rajagopalan (MS, UCSD → ML Engineer, Streamingo)

2022 - 2024

Jyothir S V (MS, NYU → RA, NYU)

2023 - 2024

Adrià Lopez (MS, ETH → PhD, ETH)

2024 - 2025

Varun Giri (MS, Georgia Tech)

2024 - 2025

Nathaniel Thomas (MS, UCSD → ML Engineer, Meta AI)

2024 - 2026

Yutao Xie (MS, UCSD → PhD, UC Santa Barbara)

2024 - 2026

Jiajun Xi (MS, UCSD)

2024 - 2026

Xueqian Bai (MS, UCSD)

2025 - 2026

Arth Shukla (BS, UCSD → PhD, Univ. Washington)	2026 - 2026
Anastasiia Pedan (MS, U. of Alberta)	2025 -
Darrien McKenzie (PhD, UCSD)	2025 -
Harshit Dhankhar (MS, UCSD)	2026 -
Madhoolika Chodavarapu (MS, UCSD)	2026 -
Zavier Andrianarivo (BS, NYU)	2026 -

Invited Talks

Sony AI	"Improving agents via online interaction"	May 2026
Univ. Michigan	"Robots that learn by doing"	Apr 2026
UC Riverside	"AI agents should learn by doing"	Apr 2026
CMU Intent Lab	"Improving agents via online interaction"	Feb 2026
U. Washington	"Improving agents via online interaction"	Feb 2026
Johns Hopkins U.	"Improving agents via online interaction"	Feb 2026
Stanford REAL	"Improving agents via online interaction"	Feb 2026
NYU CILVR	"Improving agents via online interaction"	Feb 2026
UC Berkeley RLL	"Building Massively Multitask World Models"	Jan 2026
EWM @ NeurIPS	"Massively Multitask World Models for Continuous Control"	Dec 2025
RIWM @ ICCV	"Massively Multitask World Models for Continuous Control"	Oct 2025
Cohere Labs	"World Models on an Academic Budget"	Aug 2025
CU Hong Kong	"Learning World Models for Embodied AI"	May 2025
World Models @ ICLR	"World Models on a Budget"	Apr 2025
UIUC CSLSC	"Learning World Models for Robots"	Feb 2025
KEA Copenhagen	"Data-Driven Algorithms for Robotics"	Dec 2024
USC	"Data-Driven World Models for Robots"	Oct 2024
HuggingFace	"TD-MPC, TD-MPC2, and beyond"	Jun 2024
BeNeRL Seminar	"Data-Driven World Models for Robots"	Jun 2024
TILOS Institute	"Large Datasets and Models for Robots in the Real World"	May 2024
Univ. Michigan	"Robot Learning with (Generalist) World Models"	Jan 2024
Georgia Tech	"Building Generalist World Models"	Jan 2024
Tech. Univ. Denmark	"Data-Driven World Models at Scale: Why, What, and How?"	Dec 2023
Tsinghua IIIS	"The Next Generation of World Models"	Mar 2023
MILA/ServiceNow	"World Models with Behavioral Priors"	Feb 2023
Georgia Tech	"Towards Sample-Efficient Robot Learning with World Models"	Jan 2023
Meta AI (FAIR)	"Pretraining for Control: Current Challenges and Solutions"	Jan 2023
TU Delft	"A Path Towards Generalist Agents"	Oct 2022
Generally Intelligent	Podcast: https://generallyintelligent.com/podcast/2022-12-16-podcast-episode-25-nicklas-hansen/	Sep 2022
Intel AI	"Temporal Difference Learning for Model Predictive Control"	Apr 2022
Intel AI	"Robots that Generalize"	Aug 2021
G-Research	"Agents that Generalize and Adapt"	Feb 2021

Academic Service

Workshop organizer

Workshop on Bimanual Dexterous Tool Use (CoRL)	2026
What Makes a Pretrained Policy Adaptable? (CoRL)	2026
2nd Workshop on World Models (ICLR)	2026

Journal reviewer

Robotics and Autonomous Systems (RAS)	2026
IEEE Transactions on Robotics (T-RO)	2024
Transactions on Pattern Analysis and Machine Intelligence (TPAMI)	2024
Journal of Machine Learning Research (JMLR)	2023
International Journal of Computer Vision (IJCV)	2023

Conference reviewer

Conference on Neural Information Processing Systems (NeurIPS)	2022, 2023, 2024, 2025, 2026
International Conference on Machine Learning (ICML)	2022, 2023, 2025, 2026
International Conference on Learning Representations (ICLR)	2024, 2025, 2026
Conference on Computer Vision and Pattern Recognition (CVPR)	2022, 2023, 2026
European Conference on Computer Vision (ECCV)	2022, 2026
International Conference on Computer Vision (ICCV)	2023
IEEE Robotics and Automation Letters (RA-L)	2021, 2022, 2023, 2024, 2025
IEEE International Conference on Robotics & Automation (ICRA)	2023, 2024, 2026
Annual Conference on Robot Learning (CoRL)	2025, 2026
International Conference on Intelligent Robots and Systems (IROS)	2023, 2025, 2026
Association for the Advancement of Artificial Intelligence (AAAI)	2021

Workshop reviewer

Workshop on X-Embodiment Robot Learning @ CoRL	2024
Foundation Models for Decision-Making, Workshop @ NeurIPS	2023
Self-Supervised Learning - Theory and Practice, Workshop @ NeurIPS	2023
Learning Dexterous Manipulation, Workshop @ RSS	2023
Structural and Compositional Learning on 3D data, Workshop @ CVPR	2023
Self-Supervised Learning - Theory and Practice, Workshop @ NeurIPS	2022
Generalizable Policy Learning in the Physical World, Workshop @ ICLR	2022

Workshop Presentations

Generalizable Robotic Insertion with World Models

Poster

Workshop on Out-of-Distribution Generalization in Robotics @ RSS

2025

Multi-Stage Manipulation with Demonstration-Augmented Reward, Policy, and World Model Learning

Poster

Workshop on Out-of-Distribution Generalization in Robotics @ RSS

2025

Musculoskeletal Control with TD-MPC2

Poster

MyoSuite Symposium @ NeurIPS (2nd place Locomotion track + Physiological Award)

2024

TD-MPC2: Scalable, Robust World Models for Continuous Control

Poster

Foundation Models for Decision-Making @ NeurIPS

2023

Robot Learning @ NeurIPS

2023

Pre-Training Robot Learning @ CoRL

2023

On Pre-Training for Visuo-Motor Control: Revisiting a Learning-from-Scratch Baseline

Poster

Pre-Training Robot Learning @ CoRL

2022

On the Feasibility of Cross-Task Transfer with Model-Based Reinforcement Learning

Poster

Pre-Training Robot Learning @ CoRL

2022

Foundation Models for Decision Making @ NeurIPS

2022

Deep RL Workshop @ NeurIPS

2022

MoDem: Accelerating Visual Model-Based Reinforcement Learning with Demonstrations

Poster

Pre-Training Robot Learning @ CoRL

2022

Deep RL Workshop @ NeurIPS

2022

Look Closer: Bridging Egocentric and Third-Person Views with Transformers for Robotic Manipulation

Poster

Workshop on Deployable Decision Making in Embodied Systems @ NeurIPS

2021

Deep RL Workshop @ NeurIPS

2021

Learning Vision-Guided Quadrupedal Locomotion End-to-End with Cross-Modal Transformers	<i>Poster</i>
Deep RL Workshop @ NeurIPS	2021
Visual Learning and Reasoning for Robotics Workshop @ RSS	2021
Stabilizing Deep Q-Learning with ConvNets and Vision Transformers under Data Augmentation	<i>Poster</i>
Unsupervised RL Workshop @ ICML	2021
Visual Learning and Reasoning for Robotics Workshop @ RSS	2021
Self-Supervised Policy Adaptation During Deployment	<i>Poster</i>
Microsoft Research RL Day	2021
Deep RL Workshop @ NeurIPS	2020
Workshop on Robot Learning @ NeurIPS	2020

Awards and Scholarships

2026	UC Riverside Future Academics & Mentorship Excellence (FAME) Program	<i>Award</i>
	· A 2-day workshop for future faculty applicants (8 recipients nation-wide).	
2024	NVIDIA Graduate Fellowship	<i>Fellowship</i>
	· An award of \$60,000 to cover stipend and tuition for 1 year (10 recipients worldwide).	
2023	NeurIPS Top Reviewer	<i>Award</i>
	· Awarded free conference registration as a top peer reviewer for the conference.	
2021	Robotics Summer School Scholarship	<i>Scholarship</i>
	· A scholarship to participate in a two-week summer program in Denmark.	
2020	Spar Nord Fond Scholarship	<i>Scholarship</i>
	· A scholarship to study a semester at UC Berkeley (5 recipients nation-wide).	
2020	UC Berkeley's SCET Collider Cup Finalist	<i>Award</i>
	· Biannual startup competition. Best student project from each class is nominated.	
2017	Otto Mønsted Fonds Legat	<i>Scholarship</i>
	· A grant for students with a GPA ≥ 8.0 who wish to study a semester abroad.	

Volunteering

2026	UC San Diego, CSE PhD Visit Days Organizer	<i>Organizing</i>
	· Co-organizing visit days for prospective PhD students in the CSE department.	
2025	NeurIPS, Volunteer	<i>Organizing</i>
	· Volunteer work for NeurIPS 2025 in San Diego.	
2024	Queer in AI, Graduate Application Financial Assistance	<i>Organizing</i>
	· Reviewed applications for microgrants associated with graduate applications.	
2024	UC San Diego, Graduate Women in Computing Mentor	<i>Mentorship</i>
	· Mentor for first-year PhD students that identify with a gender minority.	
2023	UC San Diego, GradAMP Mentor (PhD Applications)	<i>Mentorship</i>
2022	· Supported prospective students through weekly mentor-mentee meetings in Fall.	

Misc. Open-Source Projects

Dreamer 4 Unofficial Implementation (★380)	2026
· Unofficial PyTorch implementation of the Dreamer 4 world model.	
https://github.com/nicklashansen/dreamer4	
TD-MPC2 Official Implementation (★919)	2024
· Public code release for "TD-MPC2: Scalable, Robust World Models for Continuous Control".	
https://github.com/nicklashansen/tdmpc2	

TD-MPC Official Implementation (★523)

· Public code release for "Temporal Difference Learning for Model Predictive Control".
<https://github.com/nicklashansen/tdmpc>

2022